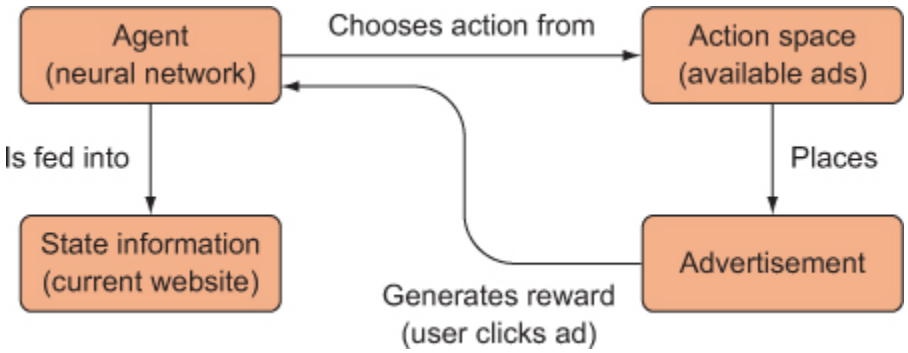
some information about that user's preferences and could help guide our decision about which ad to place. We call this contextual information a *state* and this new class of problems *contextual* bandits (see figure 2.5).

Figure 2.5. Overview of a contextual bandit for advertisement placement. The agent (which is a neural network algorithm) receives state information (in this case, the current website the user is on), which it uses to choose which of several advertisements it should place at the checkout step. Users will click on the advertisement or not, resulting in reward signals that get relayed back to the agent for learning.



Definition

A *state* in a game (or in a reinforcement learning problem more generally) is the set of information available in the environment that can be used to make decisions.

2.3.2. States, actions, rewards

Before we move on, let's consolidate some of the terms and concepts we've introduced so far. Reinforcement learning algorithms attempt to model the world in a way that computers can understand and calculate. In particular, RL algorithms model the world as if it merely involved a set of *states*, *S* (state space), which are a set of features about the environment, a set of *actions*, *A* (action space), that can be taken in a given state, and *rewards*, *r*, that are given for taking an action in a specific state. When we speak of taking a particular action in a particular state, we often call it a *state-action pair* (*s*, *a*).